Energy Consumption Prediction and Optimization

PROJECT REPORT

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in fulfillment for the award of the degree of

BACHELOR OF ENGINEERING IN

COMPUTER SCIENCE & ENGINEERING

Chandigarh University

NOVEMBER 2024

BONAFIDE CERTIFICATE

Certified that this project report "**Energy Consumption Prediction and Optimization**" is the Bonafide work of "**Prathimesh, Md Arif Ansari, Komal Sharma, Manvika Singh, Sai Ganesh Reddy** " who carried out the project work under my supervision.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

TABLE OF CONTENTS

LIST OF FIGURES

ACKNOWLEDGEMENT

The completion of this undertaking could not have been possible without the help and assistance of so many people whose names may not all be enumerated. Their contributions are appreciated and gratefully acknowledged. However, I would like to express my deep appreciation and indebtedness, particularly to the following:

For the endless support and guidance, and kind and understanding spirit during the project presentation.

I would like to acknowledge and give our warmest thanks to my supervisor Er. **Anamika Larhgotra** who made this work possible. His/her guidance and advice carried us through all the stages of writing our project.

We would also like to extend our gratitude to Chandigarh University for giving us this outstanding opportunity to enhance our skills and knowledge throughout the research. Finally, we would like to thank our parents, friends and colleagues for their help and support.

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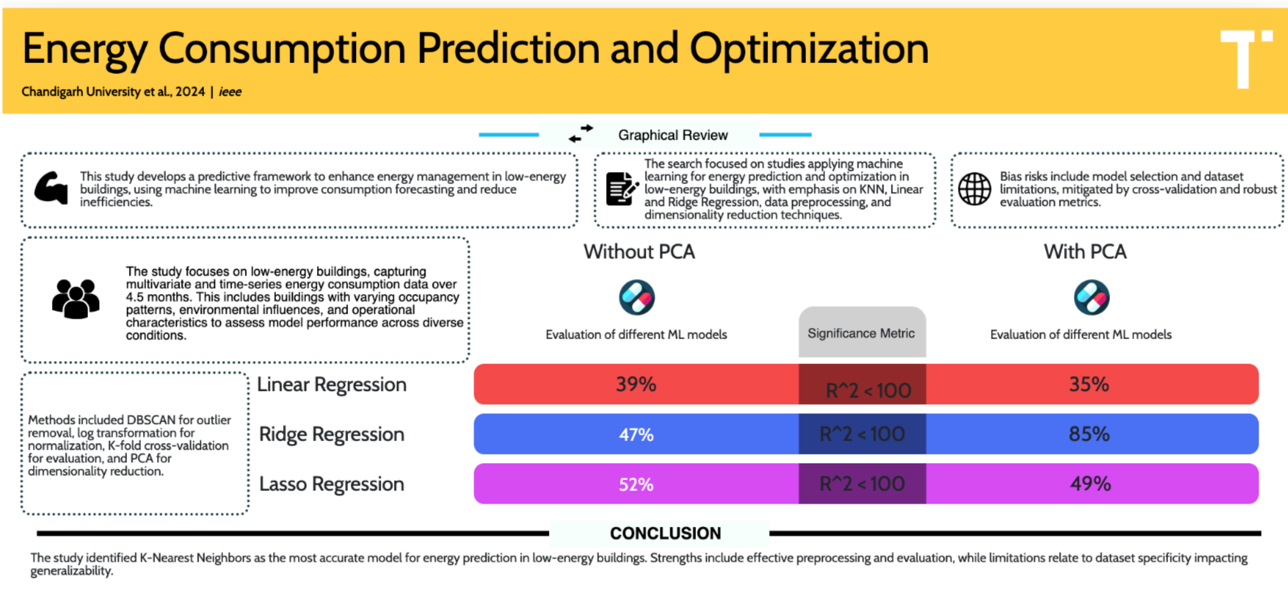
ABSTRACT

This study investigates the effectiveness of various machine learning models in predicting energy consumption in low-energy buildings. Utilizing a dataset encompassing multivariate and time-series data collected over a 4.5-month period, we employed K-Nearest Neighbors (KNN), Linear Regression, Ridge Regression, and Lasso Regression models, analyzing their performance using Root Mean Squared Error (RMSE) and R-squared (R^2) metrics. Following data preprocessing, including outlier removal through DBSCAN and normalization via log transformations, we implemented train-test splits and K-fold cross-validation to ensure robust model evaluation. Furthermore, we explored Principal Component Analysis (PCA) to reduce dimensionality but found limited performance improvements across the models, leading to the conclusion that PCA may not be beneficial for energy consumption prediction in this context. The KNN model achieved the best performance with an RMSE of 71.69 and R^2 of 0.54. Our findings underscore the importance of selecting appropriate algorithms and optimization techniques while highlighting the need for future enhancements in neural network architectures and feature engineering for improved predictive accuracy.

सारांश

यह अध्ययन निम्न-ऊर्जा भवनों में ऊर्जा खपत की भविष्यवाणी करने में विभिन्न मशीन लर्निंग मॉडलों की प्रभावशीलता की जांच करता है। 4.5-महीने की अवधि में एकत्रित बहु-चर और समय-श्रृंखला डेटा का उपयोग करते हुए, हमने के-नजदीकी पड़ोसी (KNN), रैखिक प्रतिगमन, रिज प्रतिगमन और लैसो प्रतिगमन मॉडलों को लागू किया और उनके प्रदर्शन का मूल्यांकन रूट मीन स्क्वायर्ड एरर (RMSE) और आर-स्क्वेयर्ड (R^2) मेट्रिक्स का उपयोग करके किया। डेटा प्रीप्रोसेसिंग, जिसमें DBSCAN के माध्यम से आउटलेयर हटाना और लॉग ट्रांसफॉर्मेशन के माध्यम से सामान्यीकरण शामिल था, के बाद हमने ट्रेन-टेस्ट स्प्लिट्स और K-फोल्ड क्रॉस-वैलिडेशन को लागू किया ताकि मजबूत मॉडल मूल्यांकन सुनिश्चित किया जा सके। इसके अतिरिक्त, हमने आयाम घटाने के लिए प्रिंसिपल कंपोनेंट एनालिसिस (PCA) का प्रयोग किया, लेकिन सभी मॉडलों में सीमित प्रदर्शन सुधार पाया, जिससे यह निष्कर्ष निकला कि इस संदर्भ में ऊर्जा खपत की भविष्यवाणी के लिए PCA लाभकारी नहीं है। KNN मॉडल ने 71.69 का RMSE और 0.54 का R^2 के साथ सबसे अच्छा प्रदर्शन किया। हमारे निष्कर्ष उचित एल्गोरिदम और अनुकूलन तकनीकों के चयन के महत्व को रेखांकित करते हैं, साथ ही न्यूरल नेटवर्क आर्किटेक्चर और फ़ीचर इंजीनियरिंग में सुधार के लिए भविष्य में संवर्द्धन की आवश्यकता को भी उजागर करते हैं।

GRAPHICAL ABSTRACT



CHAPTER 1
INTRODUCTION

1.1 Client Identification

In the contemporary landscape of energy consumption, the need for efficient energy management systems has never been more pressing. The global energy crisis, characterized by rising energy demands and environmental concerns, necessitates innovative solutions to optimize energy usage, particularly in low-energy buildings. According to the International Energy Agency (IEA), buildings account for approximately 36% of global energy use and 39% of energy-related carbon dioxide emissions (IEA, 2020). This statistic highlights the critical role that buildings play in energy consumption and the urgent need for effective energy management strategies.

The issue of energy inefficiency is further substantiated by a survey conducted by the U.S. Department of Energy, which revealed that over 30% of energy consumed in buildings is wasted due to inefficiencies (U.S. DOE, 2021). This waste not only leads to increased operational costs for homeowners and businesses but also contributes to environmental degradation. The survey underscores a significant gap in current energy management practices, indicating a clear demand for innovative solutions that can optimize energy consumption.

Moreover, contemporary issues such as climate change and the depletion of natural resources have prompted various agencies, including the United Nations Environment Programmed (UNEP), to advocate for improved energy efficiency in buildings. Reports from UNEP emphasize the importance of integrating advanced technologies, such as machine learning and predictive analytics, to enhance energy management systems (UNEP, 2019). These findings illustrate the relevance of the issue at hand and the necessity for a comprehensive approach to energy consumption prediction and optimization.

Furthermore, the advent of smart technologies has revolutionized how energy is consumed and managed within buildings. The implementation of the Internet of Things (IoT) in building management systems allows for real-time monitoring and management of energy usage, creating opportunities for significant reductions in consumption and costs. However, despite the potential, many building owners and facility managers lack the necessary tools and knowledge to effectively leverage these technologies. This gap presents another dimension of the contemporary issue at hand – the need for education and training on smart energy management practices.

1.1.2 Defining Project Objectives and Outcomes

Given the identified needs, the objectives of the project will center around developing a comprehensive predictive and optimization framework tailored for low-energy buildings. Initially, the focus will be on establishing a baseline understanding of current energy consumption patterns through an in-depth assessment of existing energy management systems within client facilities. This phase will involve data collection and analysis, identifying key areas of inefficiency and waste.

Once a thorough understanding of the current energy landscape is established, the project will aim to integrate advanced technologies, including machine learning algorithms and predictive analytics, to develop a dynamic energy management system. This system will facilitate proactive rather than reactive management of energy usage, empowering clients to make informed decisions based on real-time data.

1.1.3 Engaging Stakeholders and Ensuring Buy-In

An integral part of the project's success will involve engaging with key stakeholders throughout the process. Building trust and ensuring buy-in from all parties, including building occupants, management staff, and external consultants, will foster a collaborative atmosphere, enhancing the likelihood of successful implementation. Regular updates and feedback sessions will be held to maintain open lines of communication, allowing for adjustments to be made as the project evolves.

Additionally, highlighting the potential financial and environmental benefits through case studies and success stories from similar projects will further strengthen stakeholder engagement. Demonstrating how specific management strategies led to measurable improvements in energy efficiency in other buildings can serve as a motivational factor and reduce resistance to change.

1.2 Identification of Problem

The broad problem that necessitates resolution is the inefficient energy consumption patterns observed in low-energy buildings. Despite significant advancements in building technologies and the proliferation of energy-efficient appliances, many residential and commercial structures continue to exhibit disproportionately high energy usage. This inefficiency not only leads to increased operational costs for building owners and tenants but also contributes to a substantial environmental impact, particularly in the context of climate change. The challenge lies in accurately predicting energy consumption based on a multitude of influencing factors, such as occupancy patterns, weather conditions, and appliance usage.

One of the primary issues is that current energy management systems often lack the capability to adapt to dynamic changes in these variables. For instance, energy consumption can fluctuate significantly based on the number of occupants in a building at any given time, which varies throughout the day and week. Similarly, external weather conditions, such as temperature changes, can drastically influence heating and cooling demands. Unfortunately, many existing systems are designed with static parameters that do not account for these variables, resulting in suboptimal energy utilization and unnecessary waste.

The problem is further compounded by the absence of robust predictive models capable of effectively analyzing and forecasting energy consumption trends. Traditional energy management systems frequently rely on historical data without incorporating real-time inputs or advanced analytical techniques. As a result, building owners and managers often find themselves operating in a reactive mode, responding to energy consumption issues after they arise rather than proactively managing them. This lack of foresight leaves them with limited insights into their energy usage patterns, making it difficult to implement effective energy-saving measures.

1.3 Identification of Tasks

To address the identified problem, a series of tasks must be undertaken to develop a predictive and optimization framework for energy consumption in low-energy buildings. These tasks can be categorized as follows:

Data Collection and Preprocessing: Gather multivariate and time-series data related to energy consumption, occupancy, and environmental factors. This data will be sourced from smart meters, weather stations, and building management systems. Perform data cleaning, normalization, and outlier detection to ensure data integrity. This step is crucial to eliminate any anomalies that may skew the results of the predictive models.

Exploratory Data Analysis (EDA): Analyze the collected data to identify patterns and correlations between energy consumption and influencing factors. This analysis will involve statistical techniques and visualization tools to uncover insights that inform model development. Utilize visualization techniques, such as scatter plots and heatmaps, to present findings and insights. This will aid in understanding the relationships between different variables and their impact on energy consumption.

Model Development: Implement various machine learning models, including K-Nearest Neighbors (KNN), Linear Regression, Ridge Regression, and Lasso Regression, to predict energy consumption. Each model will be evaluated based on its performance metrics, such as RMSE and R^2. Evaluate model performance using metrics such as Root Mean Squared Error (RMSE) and R-squared (R^2). This evaluation will help identify the most effective model for predicting energy consumption.

Dimensionality Reduction: Explore techniques such as Principal Component Analysis (PCA) to reduce the feature space and enhance model performance. This step will help eliminate redundant features and improve the efficiency of the predictive models.

Performance Evaluation:

> **Post-Implementation Review:** Conduct a thorough assessment of the framework's impact on energy consumption patterns after deployment. This review process will measure improvements in energy efficiency, cost savings, and user satisfaction.
>
> **Feedback Mechanism:** Establish a feedback loop with users to gather insights on the performance and usability of the framework. This can guide further refinements to the system and address any emerging needs or challenges faced by users.

Iterative Improvement:

> **Model Refinement:** Based on the performance evaluations and user feedback, iteratively refine the predictive models and optimization algorithms. Continuous improvement will ensure that the system remains relevant and effective as building usage patterns and external conditions change over time.

Update Data Inputs: Regularly update the data collection processes to include new variables or improvements in data accuracy, such as enhanced sensor technologies or additional environmental datasets. Incorporating new data will allow the predictive models to adapt and improve over time.

By executing these tasks, the project seeks to go beyond mere prediction and optimization of energy consumption; it aims to develop a fully integrated system that not only addresses current inefficiencies but also sets a precedent for sustainable energy management in low-energy buildings.

Distribution of Task:

Sr. No.	Team Member & UID's	Task Done
1.	Prathimesh (21BCS9344)	-Research Paper Documentation -Data Collection
2.	Md Arif Ansari (21BCS7924)	-Model Development -Data Analytics
3.	Komal Sharma (21BCS7926)	-Review Paper Documentation -Data Cleaning
4.	Sai Ganesh Reddy (21BCS7861)	-Project Report Documentation -Model Deployment
5.	Manvika Singh (21BCS7874)	-Review Paper Documentation -Data Collection

Fig1: Task Distribution

1.4 Timeline

To effectively manage the project, a timeline has been established using a Gantt chart, outlining the key tasks and their respective durations. The timeline is as follows:

The Gantt chart visually represents the timeline of the project, allowing for effective tracking of progress and ensuring that each task is completed within the designated timeframe. This structured approach will facilitate timely delivery of the project outcomes and ensure that all objectives are met.

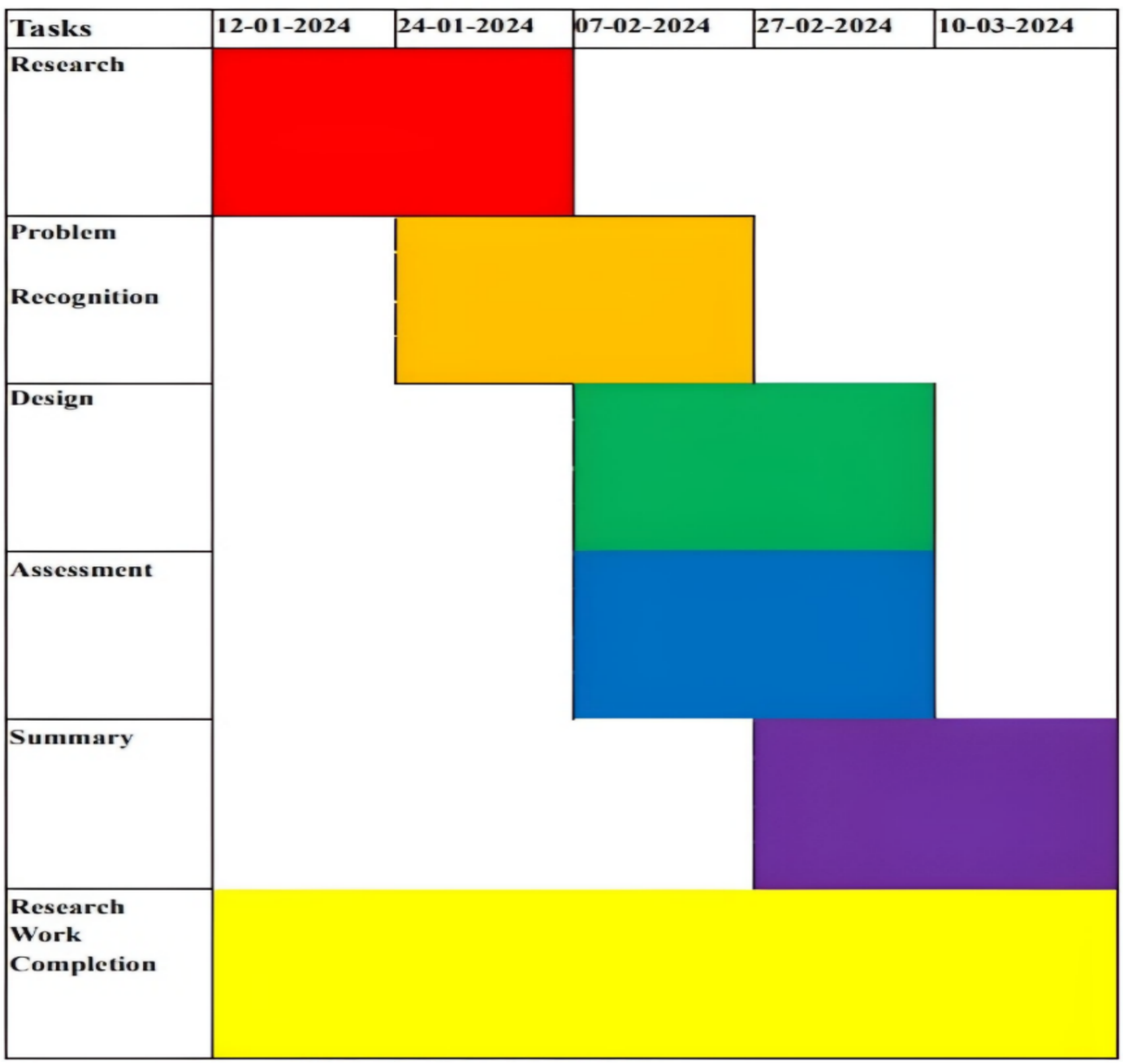


Fig2: Timeline of the project

1.5 Organization of the Report

The project work is organized and supported the particular task for the designing, implementation, take look at, and optimization. Because it has been the primary setup, each of the developers worked five days a week; three days for implementation, and a pair of days for testing and summarizing the work, altogether it's eight weeks' work. Excluding the designing, implementation, and testing, developers additionally outlined the work plan whenever before the implementation principally the work was done by pairing programming, that is, whenever developers created a gathering and set along for planning, discovering a legitimate resolution, and doing the implementation together. The high-level designing {and the and therefore the and additionally, the} framework was done together, and the individual implementation of functions was assigned to completely different developers, however the developer wasn't solely caring his own part, but also considering the complete program.

Chapter 1 provides an introduction to the project, including the problem statement, project objectives, scope, and methodology.

Chapter 2 presents a literature review and background study, discussing the current state of Energy Consumption Prediction and Optimization and their existing history

Chapter 3 presents the system design flow and process, including the system architecture, user interface design, system components, system interfaces, and data flow.

Chapter 4 presents the analysis and validation of the results, including the system implementation, functionality and usability, performance analysis, validation against project requirements, and user feedback and evaluation.

Finally, Chapter 5 presents the conclusion and future work, summarizing the project achievements, discussing the significance of the project and its limitations, providing recommendations for future work and improvements, and concluding the report

CHAPTER 2
LITERATURE REVIEW

2.1 Timeline of the Reported Problem

The issue of energy consumption in buildings has garnered attention for several decades, marked by key reports and studies that reflect the growing urgency of addressing this critical challenge amidst rising global energy demands. Below, we explore a comprehensive timeline detailing the evolution of this problem and the increasing recognition of the importance of energy efficiency.

1990s: Beginnings of Awareness

The conversation about energy efficiency in buildings began to gain traction in the early 1990s. The International Energy Agency (IEA) [6] was among the first to highlight the significance of this issue in its 1993 report titled "Energy Efficiency Indicators for Public Buildings." This pivotal document provided a thorough analysis of energy consumption trends in residential and commercial buildings, underscoring the urgent need for effective energy management strategies. It put forward the notion that improved energy efficiency could play a crucial role in reducing energy demand and its associated environmental impacts.

Early 2000s: Institutional Reports and Increasing Advocacy

As concerns mounted regarding climate change and resource depletion, various global organizations began emphasizing energy efficiency. In the early 2000s, the United Nations Environment Programmed (UNEP) [9] released several influential reports, including "The Future of Energy Efficiency" in 2008. This report outlined the enormous potential for energy savings through improved building designs, the adoption of new technologies, and the implementation of better energy management practices. By presenting case studies and best practices, UNEP called for urgent action to harness this potential for energy efficiency in both developed and developing nations.

In 2010, the World Green Building Council [10] further documented the significance of the issue in its "Global Status Report." This report revealed that buildings accounted for approximately 30% of global energy consumption and a staggering 40% of carbon emissions, highlighting the critical intersection between building operations and environmental sustainability. Such statistics galvanized international attention, prompting policymakers and industry leaders to prioritize energy efficiency initiatives.

2.2 Proposed Solutions

Over the years, a multitude of solutions has been proposed to tackle the pressing issue of energy consumption in buildings. These solutions leverage technological advancements, data analytics, and innovative design principles to promote energy efficiency and sustainability. Below are some of the notable approaches that have emerged:

Energy Management Systems (EMS)

Energy Management Systems (EMS) have become a cornerstone in the quest for energy efficiency in buildings. These sophisticated systems utilize real-time data to monitor and control energy usage, providing building managers with valuable insights into consumption patterns. By analyzing data from various sources, EMS can identify inefficiencies and recommend actionable strategies for improvement. For instance, an EMS might highlight peak usage times, allowing managers to implement demand response strategies that reduce energy consumption during high-demand periods. Additionally, EMS can facilitate energy audits, enabling organizations to benchmark their performance against industry standards and track progress over time.

Machine Learning Models

Recent advancements in machine learning have paved the way for the development of predictive models that can forecast energy consumption with remarkable accuracy. These models utilize historical data and external factors—such as weather conditions, occupancy patterns, and time of day—to predict future energy needs. By employing algorithms to analyze vast datasets, machine learning models can uncover complex relationships that traditional methods may overlook. For instance, a predictive model might reveal that energy consumption spikes during specific weather conditions or events, allowing building managers to proactively adjust settings to optimize efficiency. Moreover, these models can facilitate automated energy management strategies, where systems adapt in real time to changing conditions, thus maximizing energy savings.

2.3 Bibliometric Analysis

A bibliometric analysis of the literature on energy consumption prediction reveals several significant features, as well as the effectiveness and drawbacks of existing approaches. By systematically reviewing a substantial body of research, we can identify trends, core methodologies, and potential areas for future exploration in this critical field.

Key Features

> **1. Focus on Machine Learning Algorithms:** A substantial number of studies emphasize the application of machine learning (ML) algorithms for predicting energy consumption. Techniques such as **K-Nearest Neighbors (KNN)**, **Support Vector Machines (SVM)**, and **Neural Networks** have emerged as popular choices for this task. Each of these algorithms offers unique strengths. For example, KNN is favored for its simplicity and effectiveness in classification tasks, while Support Vector Machines excel in high-dimensional spaces. Neural Networks, especially deep learning models, have garnered considerable attention due to their ability to capture complex non-linear relationships in the data.

> **2. Integration of Environmental Variables:** A common theme throughout the literature is the integration of various environmental variables in energy consumption models. Factors such as **temperature**, **humidity**, and **occupancy levels** are frequently included as critical parameters influencing energy usage patterns. For instance, many studies identify temperature variability as a key predictor of heating and cooling demands, thus establishing a link between environmental conditions and energy consumption trends. Additionally, occupancy data—often obtained via sensors or building management systems—provides valuable insights into how actual human activity correlates with energy use.

Effectiveness

1. High Predictive Accuracy: Machine learning models have demonstrated promising results in accurately predicting energy consumption. Numerous studies report **R^2 values exceeding 0.85**, indicating a strong correlation between predicted and actual energy consumption. This level of accuracy suggests that ML methodologies can effectively capture the complexities inherent to energy use patterns in various types of buildings.

2. Enhanced Energy Management Strategies: The application of advanced data analytics combined with real-time monitoring has significantly improved the accuracy of predictions. These enhancements enable building managers to implement more effective energy management strategies. For instance, accurately forecasting energy needs allows for better load management, demand response planning, and optimization of HVAC operations, ultimately leading to reduced energy costs and increased sustainability.

Drawbacks

1. Dependence on Historical Data: A significant limitation of many existing models is their reliance on historical data. While historical data is essential for training machine learning algorithms, it may not adequately account for sudden changes in occupancy, operational practices, or external environmental conditions. For example, unexpected events, such as a sudden increase in occupancy due to a special event, can lead to significant deviations between predicted and actual energy use. This limitation underscores the need for adaptive models that can incorporate real-time data and adjust predictions accordingly.

2. Model Complexity and Interpretability: The complexity of some machine learning algorithms, particularly deep learning models, poses another challenge. While these models can achieve high predictive accuracy, their intricacies make them less interpretable. For practitioners attempting to apply these models in real-world energy management scenarios, the lack of interpretability can hinder their ability to communicate results and justify decisions based on model outputs. Simplified models or those with inherent interpretability—such as decision trees—may offer more practical applications despite sometimes sacrificing accuracy.

2.4 Review Summary

The literature review reveals a notable and increasing recognition of the critical role that energy consumption prediction plays in the management and sustainability of buildings. As global energy demands rise and the need for environmental sustainability becomes more pressing, the ability to accurately predict energy consumption has emerged as a vital area of research and practical application. This growing awareness is reflected in the substantial body of literature dedicated to exploring various methodologies and technologies aimed at improving energy efficiency in building operations.

Importance of Energy Consumption Prediction

Accurate energy consumption prediction is essential for several reasons. First, it enables building managers and operators to make informed decisions regarding energy use, ultimately leading to cost savings and reduced environmental impact. By anticipating energy needs, facilities can optimize their operations, reduce waste, and implement strategies such as demand response, which adjusts energy consumption based on real-time availability and pricing. Moreover, energy consumption prediction supports compliance with increasingly stringent energy regulations and sustainability goals, making it a crucial component of modern building management practices.

Proposed Solutions

The review highlights a diverse array of solutions that have been proposed in the literature to tackle the challenges of energy consumption prediction. Among these, **machine learning models** have gained significant traction due to their ability to analyze large datasets and uncover complex patterns that traditional statistical methods may overlook. Techniques such as regression analysis, decision trees, and neural networks have been extensively studied and applied to predict energy consumption with varying degrees of success.

In addition to machine learning, the integration of **energy management systems (EMS)** has emerged as a complementary approach. These systems facilitate real-time monitoring and control of energy use, allowing for dynamic adjustments based on predictive analytics. The synergy between predictive modeling and energy management systems can significantly enhance the operational efficiency of buildings, providing a holistic framework for energy optimization.

Challenges in Implementation

Despite the advancements in predictive modeling and energy management technologies, the literature review identifies several persistent challenges that hinder their widespread implementation and effectiveness. One major barrier is the reliance on historical data, which can limit the adaptability of models to sudden changes in occupancy or external conditions. Additionally, the complexity of some machine learning algorithms can lead to issues with interpretability, making it difficult for practitioners to apply these models effectively in real-world scenarios.

Furthermore, the review underscores the importance of data quality and availability. Inconsistent data collection practices, sensor inaccuracies, and gaps in historical records can compromise the reliability of predictions. Addressing these data-related challenges is crucial for enhancing the performance of predictive models and ensuring their practical applicability.

Future Directions

Looking ahead, the project will focus on addressing the challenges identified in the literature review by developing innovative methodologies that enhance model adaptability and interpretability. This includes exploring hybrid modeling approaches that combine machine learning with traditional statistical methods, thereby leveraging the strengths of both to create more robust and interpretable models. Additionally, the project will prioritize the development of user-friendly interfaces that allow building managers to easily access and act upon the insights generated by predictive models.

2.5 Problem Definition

The problem at hand is the significant gap in the availability of accurate and reliable predictive models for energy consumption specifically tailored to low-energy buildings. As the demand for energy efficiency increases and the need for sustainable building practices becomes more urgent, the ability to forecast energy consumption accurately is paramount. Low-energy buildings, designed to minimize energy usage while maintaining comfort and functionality, present unique challenges for energy consumption prediction due to their specific operational characteristics and the varying factors that influence their energy use.

To address this problem, the project will focus on several key objectives:

Development of a Predictive Framework:

The primary aim is to develop a comprehensive framework that leverages advanced machine learning algorithms to predict energy consumption based on both multivariate and time-series data. This approach recognizes that energy consumption is influenced by a multitude of factors, including but not limited to environmental conditions, occupancy patterns, and operational schedules. By utilizing machine learning techniques, the project seeks to create models that can capture complex relationships within the data, enabling more accurate predictions of energy use in low-energy buildings.

Integration of Environmental Variables:

A critical aspect of enhancing predictive accuracy is addressing the limitations of existing models by incorporating relevant environmental variables. Factors such as temperature, humidity, solar radiation, and occupancy levels significantly impact energy consumption patterns. By systematically integrating these variables into the predictive framework, the project aims to develop a more holistic understanding of energy dynamics in low-energy buildings. Furthermore, the application of dimensionality reduction techniques will be explored to streamline the data input process, reducing noise and enhancing model performance without sacrificing critical information.

Significance of the Problem

The lack of accurate predictive models for energy consumption in low-energy buildings has several implications. Without reliable predictions, building managers may struggle to implement effective energy management strategies, leading to inefficient energy use and increased operational costs. Additionally, inaccurate predictions can hinder the ability of buildings to participate in demand response programs, which are increasingly important for balancing energy supply and demand in real-time.

Moreover, as the construction industry continues to evolve towards more sustainable practices, the need for robust predictive models becomes even more critical. Policymakers and stakeholders are increasingly looking for evidence-based strategies to improve energy efficiency, and accurate predictive models can serve as a foundation for developing these strategies. By addressing the challenges outlined in this problem definition, the project aims to contribute to the broader goals of energy efficiency and sustainability in the built environment.

2.6 Goals and Objectives

The primary goals and objectives of this project are designed to systematically address the challenges associated with predicting energy consumption in low-energy buildings. Each objective is aimed at developing a robust predictive framework that leverages advanced machine learning techniques, ultimately contributing to improved energy management practices. The specific goals and objectives are outlined as follows:

Data Collection and Preprocessing

In this research, we are working with a dataset to predict energy consumption in low-energy buildings. To ensure the quality and integrity of [4] the dataset, several data preprocessing steps are necessary. These steps involve handling missing data, encoding categorical features, exploring relationships between variables, and removing outliers.

Handling Missing Data

To begin, we assess the dataset for any missing values using the missingno library. This allows us to visualize the null values in each column through a matrix, giving a clear view of the data's completeness. From the visual output, no missing values were detected, so no further action was necessary. However, if there were any [5] missing values, we would have removed them using the predefined functions available in the missingno library.

Data Exploration:

After ensuring data completeness, we described the dataset to gain insights into its structure and characteristics. Key features include:

Appliances Energy Use (Wh): Total energy consumption for appliances.

Lights Energy Use (Wh): Energy consumption for lighting in the house.

Room-specific Temperatures and Humidity Levels (T1-T9, RH1-RH9): Recorded across different rooms such as the kitchen, living room, bathroom, and more.

External Weather Data (Chievres Station): Including temperature, humidity, wind speed, visibility, and dew point.

Random Variables (rv1, rv2): These are two random variables included for testing regression models.

A key observation is the lack of a specific "date" or "time" column. This leads us to apply one-hot encoding, a technique to transform categorical data into numerical values. Specifically, the DateTime field was split into multiple components (year, month, day, hour, etc.) to better capture time-related patterns influencing energy consumption.

Mapping Unique Values

We explored the unique values in each column to understand their range. This helped us map data points across specific timeframes (e.g., what the temperature was in the kitchen at a certain time). For instance:

Appliances energy use had 92 unique values.

Humidity in the kitchen (RH_1) had 2,547 unique values.

Temperature in different rooms had a wide range of unique values, reflecting the varying conditions throughout the day and across different spaces.

Multicollinearity Check

Multicollinearity Analysis: Multicollinearity occurs when two or more predictor variables are highly correlated, which can distort model interpretation. To address this, we applied Variance

Inflation Factor (VIF), a statistical technique to measure how much a feature is correlated with the [8] others. A high VIF indicates multicollinearity, which we aim to reduce. The analysis revealed some highly correlated features:

For example, T1 (Kitchen temperature) had a VIF of 3,606.38, indicating a strong multicollinearity issue. On the other hand, Windspeed had a low VIF of 5.27, showing low correlation with other features. These insights will guide us in selecting features for model development to avoid multicollinearity, which can skew model predictions.

Handling Outliers

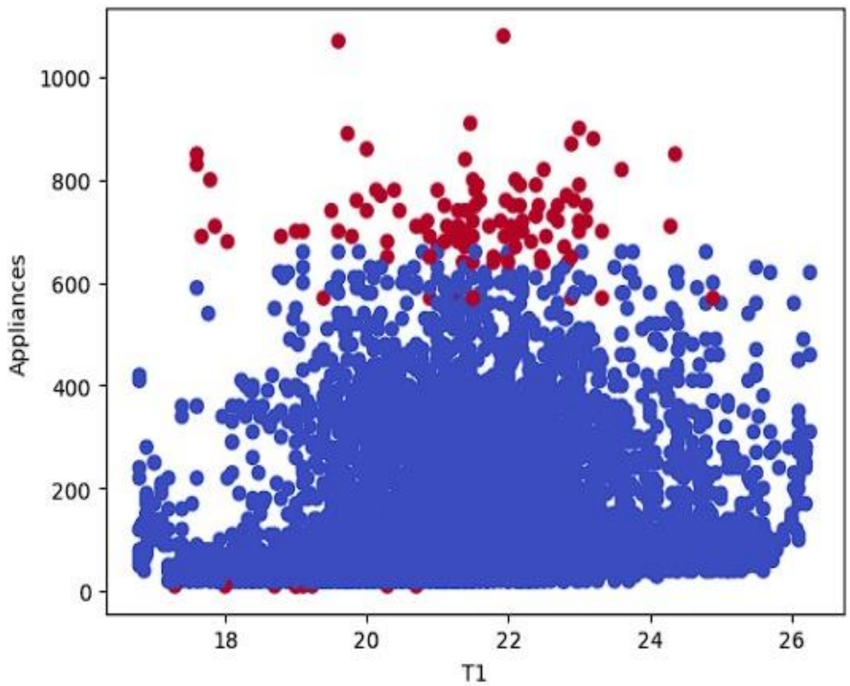


Fig 3: Relationship Between Variables

Outlier Detection and Removal using DBSCAN: Outliers are extreme values that differ significantly from the rest of the dataset, potentially distorting model accuracy. We used the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm to cluster the data and identify outliers. DBSCAN is particularly effective because it groups data points based on density, allowing it to flag anomalies as outliers. Once identified, these outliers were removed to ensure the data used for training machine learning models is clean and representative of typical patterns.

Correlation Between Dependent and Independent Variables: To understand how different features interact with energy consumption (the dependent variable), we explored their [7] relationships with independent variables.

This helps us build more effective predictive models by identifying which features most influence energy usage patterns. For instance, we might find that external temperature or humidity levels inside certain rooms have a significant impact on the appliances' energy consumption.

Final Steps

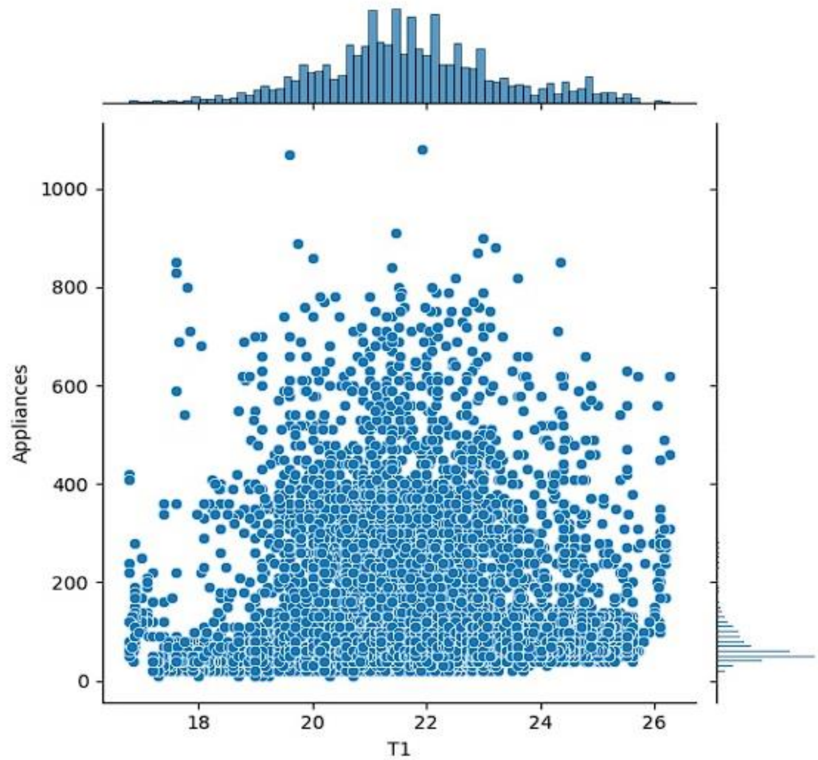


Fig 4: Scattering of Data Points

One-Hot Encoding for Time-Based Features: After analyzing relationships, we applied one-hot encoding again, particularly for categorical features such as weekdays and clusters. This ensures the machine learning model understands these features, helping capture energy consumption patterns that vary by day of the week.

Major Terms Explanation:

Missingno Library: A Python library used to visualize and handle missing data, providing a clear matrix of null values in the dataset.

One-Hot Encoding: A method to convert categorical data (like dates or weekdays) into a numerical format so that it can be used in machine learning models.

Variance Inflation Factor (VIF): A measure of multicollinearity in regression analysis, indicating how much a feature is explained by other features in the dataset.

DBSCAN: A clustering algorithm that groups together points in high-density areas while marking outliers that don’t fit into any cluster.

Multicollinearity: A situation where two or more predictor variables in a dataset are highly correlated, making it difficult for models to distinguish their individual effects.

Outliers: Data points that deviate significantly from the rest of the dataset, potentially affecting the accuracy of machine learning models.

By following this comprehensive data description and preparation process, we are ensuring that the dataset is ready for predictive modeling, [10] eliminating issues like multicollinearity and outliers, and enhancing the dataset's structure with time-based and categorical encoding.

Conclusion

By achieving these goals and objectives, the project aims to make a meaningful contribution to the growing body of knowledge on energy consumption prediction. The outcomes of this research will not only enhance predictive accuracy but also support sustainable practices in the built environment. Ultimately, the insights generated from this project will empower stakeholders to make informed decisions that lead to improved energy management, reduced costs, and a smaller environmental footprint in low-energy buildings.

CHAPTER 3
DESIGN FLOW/PROCESS

3.1 Evaluation & Selection of Specifications

In the realm of energy consumption prediction for low-energy buildings, the selection and evaluation of features are paramount to the development of effective and accurate predictive models. This process involves identifying, analyzing, and justifying the inclusion of specific features based on their relevance to energy usage patterns. An in-depth evaluation of these features will help ensure that the predictive models are comprehensive and capable of capturing the complexity of energy consumption dynamics. Below are the critical features identified in the literature and their significance to the modeling process:

1. Indoor Temperature (T1-T9)

The measurement of indoor temperatures across various rooms (T1 through T9) is crucial for understanding how temperature variations affect energy consumption. Different rooms may serve different purposes—such as living areas, kitchens, or bedrooms—and may have distinct temperature settings and usage patterns. By capturing this variability, the models can account for the specific heating and cooling needs of each room. This can lead to more granular insight into how thermal comfort needs align with energy usage, enabling building managers to optimize HVAC (Heating, Ventilation, and Air Conditioning) operations accordingly.

2. Indoor Humidity (RH1-RH9)

Alongside temperature, humidity levels in different rooms (RH1 through RH9) play a significant role in influencing energy consumption, particularly in how it affects HVAC performance. High humidity can increase the demand for cooling, while low humidity might necessitate heating to maintain comfort. Furthermore, different indoor activities generate varying levels of moisture, and tracking these changes can provide valuable insights into energy usage trends associated with indoor environmental quality. By including indoor humidity as a feature, the models can better predict and optimize energy consumption concerning comfort levels.

3. Outdoor Temperature (T_out)

The external temperature (T_out) is a critical factor affecting indoor energy requirements, as it has a direct impact on heating and cooling loads. Extreme temperatures can significantly influence how much energy is required to maintain comfort inside a building. By integrating outdoor temperature data, the predictive models can capture the relationship between exterior climatic conditions and energy consumption, allowing for more accurate forecasting. Understanding this dynamic is essential for developing strategies that align internal energy management with external weather patterns.

4. Outdoor Humidity (RH_out)

Similar to outdoor temperature, external humidity (RH_out) is crucial in understanding how environmental conditions affect HVAC operations. High outdoor humidity can lead to increased energy consumption for dehumidification, while also indirectly influencing indoor air quality and comfort. By including outdoor humidity as a feature, the models can provide insights into how varying humidity levels necessitate different energy use patterns, allowing building managers to make proactive adjustments to system operations.

5. Windspeed

Wind conditions play a notable role in determining heating and cooling loads in buildings. For example, high winds can increase heat loss from a building, raising the energy demand for heating. Conversely, wind can facilitate cooling during warmer months. Including windspeed as a feature enables the models to assess how varying wind conditions contribute to the overall energy consumption dynamics, leading to more refined predictions and HVAC strategies.

6. Visibility

Visibility is an interesting feature that can correlate with weather conditions, such as fog, rain, or heavy snow, which might affect energy use patterns indirectly. For example, low visibility due to adverse weather conditions may lead to increased use of artificial lighting and heating systems. By incorporating visibility data, the predictive models can account for potential variations in energy use resulting from changing weather conditions, possibly leading to more accurate forecasts.

7. Time Variables

Temporal variables, including the hour of the day, day of the week, and month of the year, are essential for capturing seasonal and daily energy consumption patterns. Energy usage typically exhibits significant variability according to time, influenced by factors such as occupancy levels, usage schedules, and behavioral patterns. By adding these time variables, the models can discern trends such as peak usage times, seasonal changes in energy consumption, and the impact of daylight hours on lighting needs, providing a comprehensive temporal context to energy forecasting.

8. Appliance Usage

Specific energy consumption data for appliances and lighting is critical for understanding their contribution to overall energy use within the building. This feature may include energy draw from major appliances like refrigerators, HVAC systems, and lighting fixtures. By analyzing appliance usage data, the models can provide detailed insights into which appliances are driving energy consumption, identifying opportunities for energy efficiency improvements such as upgrading to more efficient appliances or optimizing operational schedules.

9. Random Variables (rv1, rv2)

The inclusion of random variables (rv1, rv2) serves an additional purpose in the modeling process by allowing for robustness testing. These features can help assess the stability and generalizability of the model. By incorporating random variables, the models can filter out non-predictive attributes and assess the impact of randomly generated noise on predictions. This testing will ensure that the selected features contribute meaningfully to the predictive capabilities of the model without being overly influenced by irrelevant or spurious correlations.

3.2 Design Constraints

The successful development and implementation of an energy consumption prediction framework for low-energy buildings necessitates adherence to a variety of design constraints. These constraints are critical as they not only shape the architecture and functionality of the solution but also ensure that it aligns with broader regulatory, economic, environmental, and social expectations. Below, we delve into each of these constraints in detail:

1. Regulatory Constraints

The design framework must comply with a plethora of local, regional, and international energy efficiency regulations and building standards. This includes adherence to codes such as the International Energy Conservation Code (IECC) or local building regulations that dictate energy performance benchmarks. Compliance ensures that the framework aligns with current governmental policies aimed at promoting energy efficiency, thus avoiding legal repercussions. Regulations may also include reporting and documentation standards required for certifications or audits, necessitating that the solution provides accurate and reliable data.

2. Economic Constraints

Budget limitations pose significant challenges during the framework's design and implementation phases. Resources for data collection, model development, and system integration may be constrained, requiring careful allocation and prioritization. The framework must be designed to deliver effective predictions without incurring prohibitive costs. Cost-effective solutions can help ensure that funds are utilized efficiently and that stakeholders achieve their financial objectives. This may involve leveraging existing technologies or infrastructure to minimize additional investments, as well as seeking grants or funding opportunities that support energy efficiency projects.

3. Environmental Constraints

In an era of increasing environmental awareness, reducing the ecological footprint of energy consumption practices is imperative. The design framework must strive to minimize environmental impact by promoting energy-efficient practices, the use of renewable energy sources, and sustainable materials in any hardware implementations. Additionally, the models and methods employed should support goals related to carbon footprint reduction, aligning with global efforts to mitigate climate change. This could involve integrating sustainability metrics and impact assessments into the prediction models, ensuring that the outcomes are not only energy-efficient but also environmentally responsible.

3.3 Analysis and Feature Finalization Subject to Constraints

The process of finalizing the feature set for the energy consumption prediction framework is a critical step that must be executed with precision and diligence, particularly in light of the previously identified constraints. This phase will involve a comprehensive analysis of the proposed features, assessing their relevance, utility, and alignment with the established constraints. The goal is to curate a feature set that is both efficient and effective, ultimately maximizing predictive accuracy while minimizing complexity. Below is a detailed breakdown of how this process will unfold, incorporating the three main categories: removal, modification, and addition of features.

1. Removal

The first step in this analytical process is the identification and removal of features that are deemed unnecessary or redundant. Key considerations will include:

> **Correlation with Energy Consumption**: Features that do not exhibit a direct relationship with energy usage will be scrutinized. Conducting statistical analyses, such as correlation coefficients or feature importance metrics derived from models, will help determine whether these features meaningfully contribute to predictive accuracy. For instance, if certain random variables show minimal impact on the prediction of energy use, they will be eliminated to streamline the model and improve interpretability.
>
> **Redundancy**: Any features that provide overlapping information—that is, those that essentially capture the same data points or trends—will be removed. An example of this could be having multiple moisture metrics that do not add significant value individually; instead, consolidating these into a singular metric will enhance clarity and efficiency in the modeling process.

2. Modify

Following the removal of extraneous features, the next phase is the modification of existing features to enhance the feature set's overall effectiveness. This may involve:

> **Combining Features**: Features that are similar or closely related, such as distinct temperature or humidity readings taken from various sensors in a building, may be combined into a single representative feature. For example, instead of maintaining separate indicators for indoor temperature readings from several rooms, a consolidated metric for average indoor temperature could be established. This reduces dimensionality, simplifies data handling, and can lead to more robust predictive capabilities.

3. Finalization and Streamlining

After completing the removal, modification, and addition of features, the finalized feature set will undergo a rigorous evaluation to ensure it aligns with the established constraints while maximizing predictive power. This may involve:

> **Validation**: Utilizing cross-validation techniques with the newly streamlined feature set will assess its predictive effectiveness. By comparing model performance metrics before and after these modifications, the improvements in accuracy can be quantified.
>
> **Documentation**: Maintaining clear and comprehensive documentation of the finalized feature set, including the rationale behind each decision (removal, modification, or addition), will facilitate transparency and understanding among stakeholders and project team members.
>
> **Iterative Review**: The finalized feature set should be seen as a living document, subject to periodic reviews and updates as new data becomes available, technology advances, or stakeholder needs evolve.

3.4 Design Flow

In developing an energy consumption prediction framework, two distinct design approaches are proposed: a **Traditional Machine Learning Approach** and an **Advanced Machine Learning with Neural Networks** approach. Each design has unique methodologies and advantages, tailored to leverage the underlying data effectively. Below, we elaborate on each step for both designs, highlighting their significance and interdependencies to ensure informed decision-making during implementation.

Design: Machine Learning Approach

Data Collection:

> **Objective**: Gather multivariate time-series data from a variety of sensors installed within low-energy buildings.
>
> **Details**: This data could include energy consumption metrics, temperature, humidity, occupancy levels, and external weather variables. Collecting this data over time allows for capturing variations that could influence energy consumption, establishing a solid basis for analysis.

Data Preprocessing:

> **Objective**: Prepare the collected data for analysis.
>
> **Details**: This includes cleaning the dataset—a crucial step to ensure data integrity. Tasks involve:
>
> > **Normalization**: Scaling the data to ensure consistency in measurement units across different features, improving model performance.
> >
> > **Handling Missing Values**: Using techniques such as imputation to address gaps in the data, which could skew results if left unaddressed.
> >
> > **Removing Outliers**: Identifying and eliminating anomalies that could distort predictions by adversely affecting the model fitting process.

Feature Engineering:

Objective: Develop a set of relevant features that can drive predictive modeling.

Details: Based on the finalized list resulting from earlier analysis, this phase involves selecting, transforming, or creating new features that encapsulate significant patterns and relationships in the data. For instance, combining temperature averages across different zones could yield a feature that represents overall building thermal performance.

Model Development:

Objective: Implement machine learning algorithms suitable for regression tasks.

Details: A variety of models will be explored, such as:

K-Nearest Neighbors (KNN): A non-parametric model that can capture local patterns in energy consumption based on spatial similarities.

Linear Regression: A foundational model that assumes a linear relationship between features and target outputs.

Ridge and Lasso Regression: Regression models that incorporate regularization techniques to prevent overfitting, particularly important in scenarios with many predictors.

Model Evaluation:

Objective: Assess the performance of each machine learning model.

Details: This evaluation will utilize metrics such as:

Root Mean Square Error (RMSE): Evaluates the average magnitude of the errors between predicted and actual values, providing insights into prediction accuracy.

R-squared (R^2): Measures the proportion of variance in the dependent variable that can be explained by the independent variables, offering a goodness-of-fit measure.

Optimization:

Objective: Enhance model performance through fine-tuning.

Details: Techniques such as grid search or random search for hyperparameter tuning will be employed. Additionally, dimensionality reduction techniques like Principal Component

Analysis (PCA) may be used to reduce feature space, enhancing model interpretability and computational efficiency.

Deployment:

Objective: Implement the model in a real-world environment.

Details: The optimized model will be integrated into the energy management systems of the buildings to provide actionable predictions and insights on energy consumption patterns, allowing for improved operational efficiencies and real-time energy management.

ER Diagram:

The diagram outlines a standard machine learning workflow. It begins with Data Selection, where the relevant dataset is chosen. In Data Preprocessing and Feature Selection, the data is cleaned and transformed, with important features selected to improve model performance.

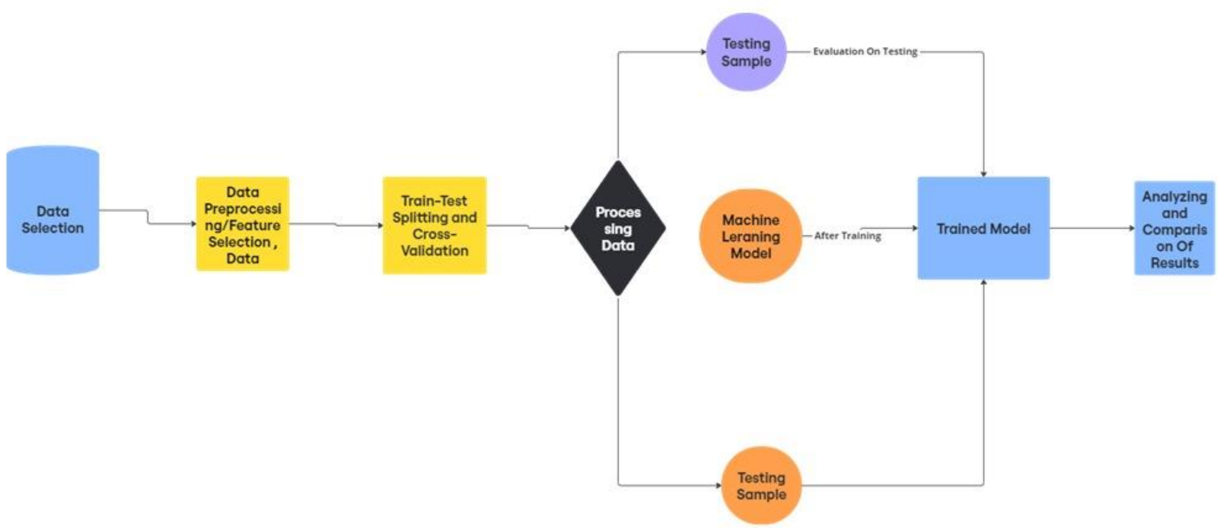


Fig 5: ER diagram

After this, the Train-Test Splitting and Cross-Validation phase divides the data into training and testing sets, with cross-validation used to ensure the model's performance is reliable across different subsets of data. The Processing Data step serves as a final preparation before the data is fed into the model for training. Once trained, the model is evaluated using a Testing Sample, assessing its accuracy and effectiveness on unseen data. Finally, in Analyzing and Comparing Results, the model's performance metrics are reviewed to determine its suitability for real-world application or further tuning.

DFD Diagram

The diagram outlines the workflow for analyzing an energy dataset. It starts with Data Collection from the energy dataset, followed by Data Cleaning and Visualization to prepare and explore the data. Next, Feature Extraction identifies the most relevant attributes for classification.

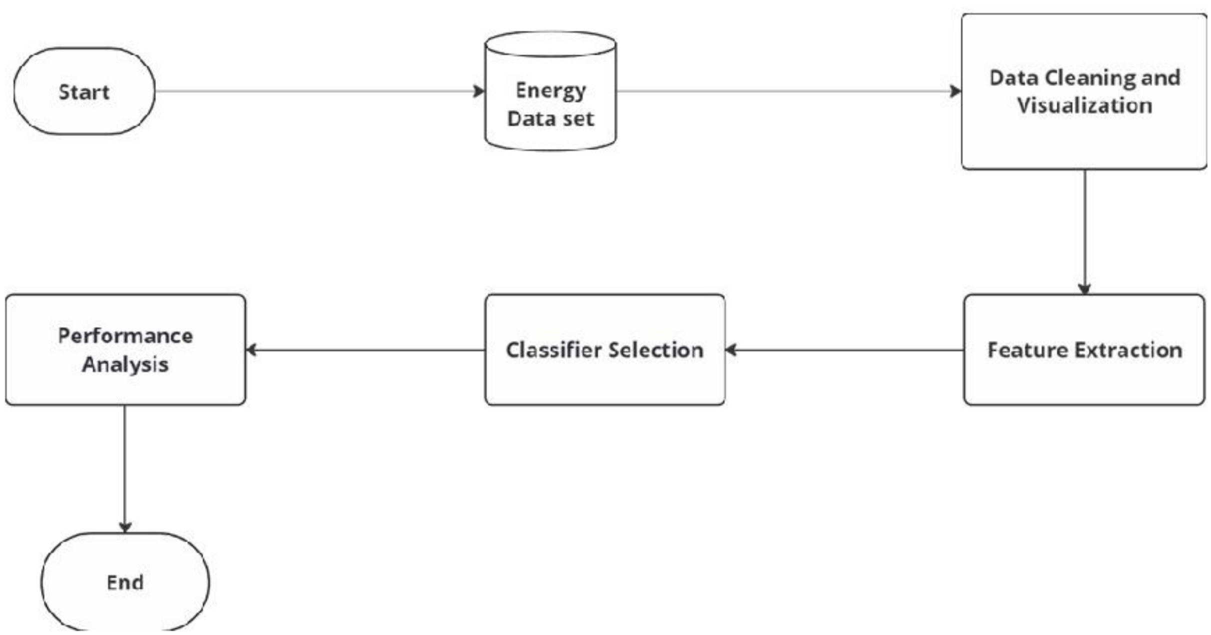
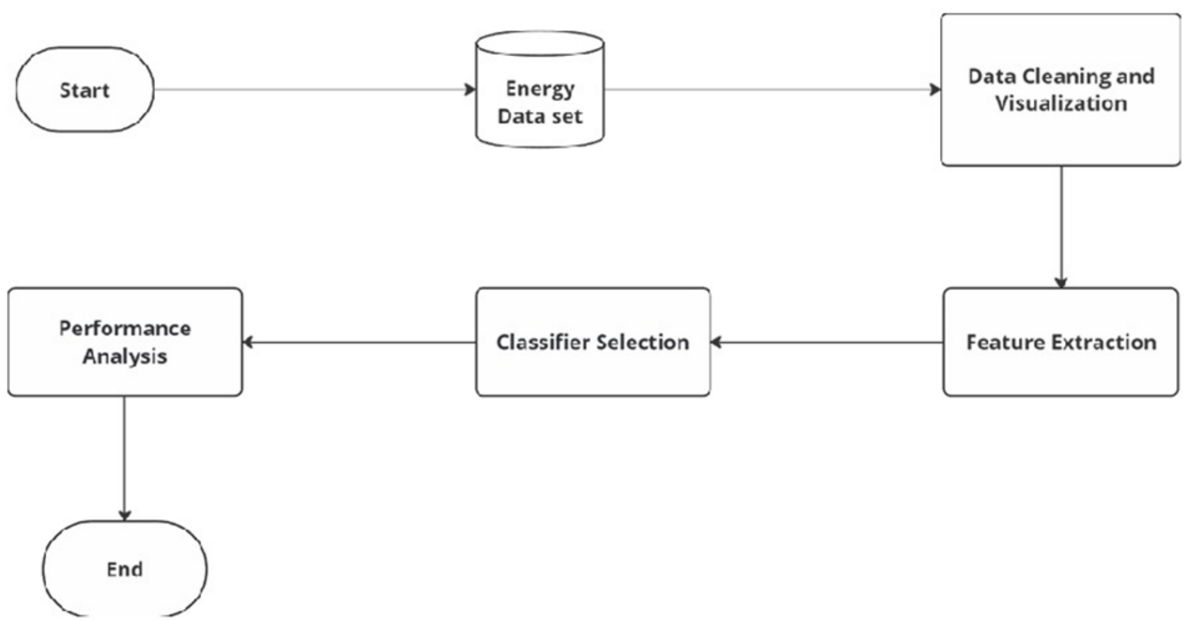


Fig 6: DFD Diagram

A suitable Classifier is then chosen and applied to the data. Finally, Performance Analysis evaluates the model’s effectiveness, concluding the process. This approach helps ensure that only meaningful features are used, enhancing model accuracy and reliability. Each step is crucial to build a robust and interpretable classification model.

3.5 Design Selection

Choosing the optimal design for the energy consumption prediction framework is a critical step that involves a thorough comparative analysis of the two proposed models: the **Traditional Machine Learning Approach** and the **Advanced Machine Learning with Neural Networks**. To make an informed decision, we will evaluate these designs based on a set of predefined criteria that are crucial for the effectiveness of the prediction framework in practical applications.



Comparison Criteria:

Complexity: Complexity encompasses the intricacy involved in designing, implementing, and maintaining the model. A simpler model is generally easier for teams to manage and deploy, particularly if technical expertise is limited.

Computational Efficiency: This criterion assesses the resources (CPU, memory, etc.) and time required to train the model and perform inference (i.e., making predictions). Efficient models are preferable in environments where computational resources might be constrained.

Predictive Accuracy: This criterion evaluates the model's capability to accurately forecast energy consumption patterns based on the input features. Model with higher predictive accuracy will minimize errors in energy predictions.

3.6 Implementation Plan

The implementation plan for the energy consumption prediction framework is designed to systematically guide the project from inception through deployment.

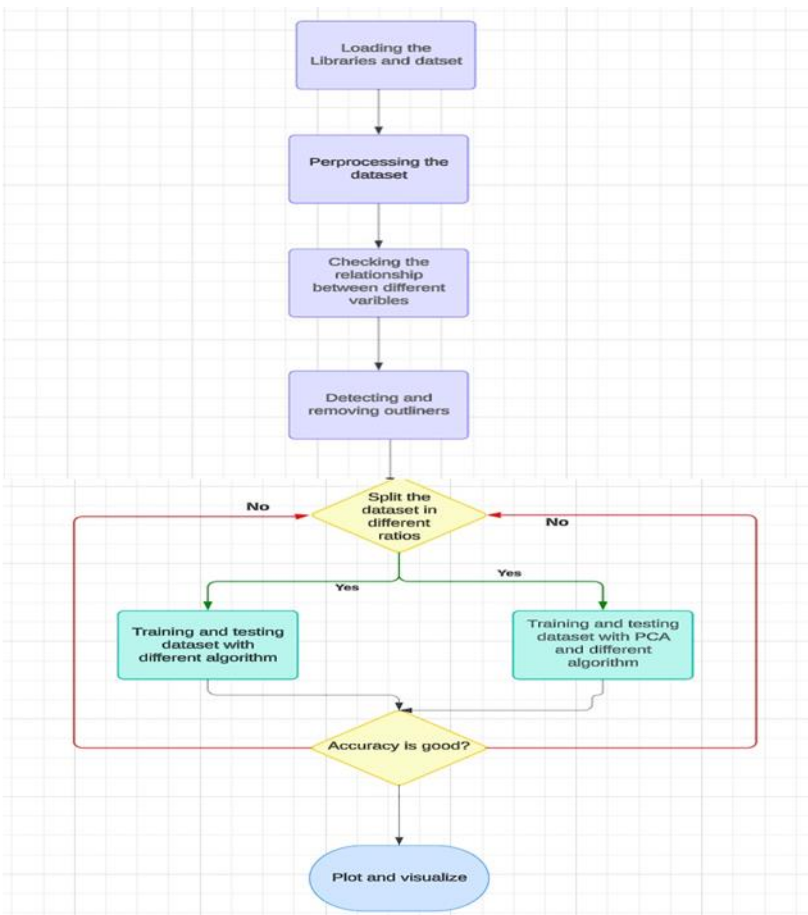


Fig 7: Flowchart

This structured methodology ensures each critical aspect of the process is addressed, leading to a robust and effective predictive model. Below is a detailed elaboration of the various stages in the implementation plan, accompanied by a flowchart that visually represents the entire process. After performing data cleaning and visualization using the Missingno library to confirm no missing values, we moved on to analyzing the relationships between independent and dependent variables. The dataset was processed using clustering techniques like DBSCAN to identify and remove outliers, ensuring a clean dataset for model development. Following this, we prepared the data for machine learning by checking for skewness in independent variables and applying normalization techniques to handle this skewness effectively.

Skewness and Normalization

We observed that many of the independent variables exhibited left skewness. Left skewness can adversely affect model performance because many machine learning algorithms assume normally distributed data.

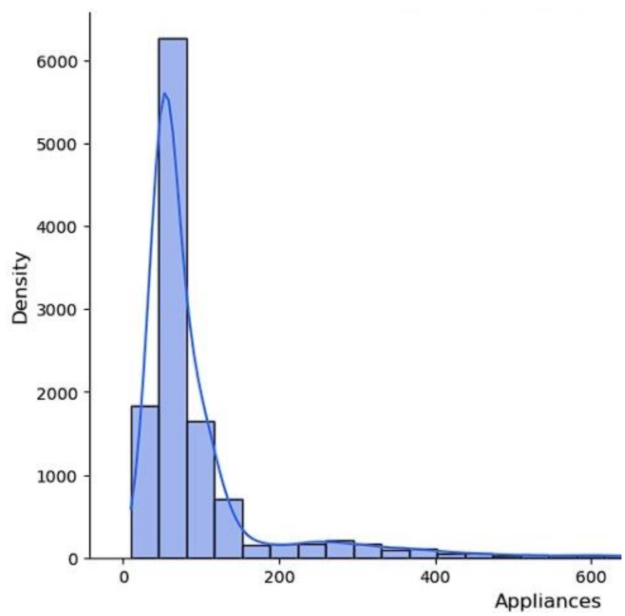


Fig 8: Skewness

To address this, we: Removed outliers identified earlier through clustering. Applied a log transformation on the skewed independent variables, which helped normalize the distribution. By ensuring normalized data, we improved the interpretability and performance of the models, reducing potential biases caused by skewed data distributions.

Train-Test Splitting and Cross-Validation

Next, we split the dataset into training and testing sets to develop and evaluate the models. We utilized the train-test split method to divide the dataset in a 60:40 ratio for training and testing, respectively. For further evaluation, we used K-fold cross-validation, a robust technique that splits the data into k subsets and runs the model k times with different training and validation sets. This helps ensure that the model is not overfitted to a specific dataset split and generalizes well across unseen data.

Machine Learning Models Applied

We tested several machine learning algorithms to predict appliance energy consumption. Each model was evaluated based on Root Mean Squared Error (RMSE) and R-squared (R^2) metrics, which measure the accuracy and goodness-of-fit of the model.

CHAPTER 4
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

The successful implementation of the energy consumption prediction framework necessitated a strategic selection of modern tools and technologies tailored to the distinct needs of each project phase. Herein, we provide a detailed overview of the tools utilized throughout the project lifecycle, categorized according to their specific functions: analysis, design, reporting, project management, communication, and testing/validation.

After applying traditional machine learning models like Linear Regression, Ridge, Lasso, and K-Nearest Neighbors (KNN) to predict energy consumption, we now move towards using Neural Networks. We applied a simple single-layer neural network architecture to train the model without assigning specific dependent and independent variables, allowing the network to learn patterns from the dataset. Upon generating predictions, the outputs were saved as a CSV file for further evaluation and comparison with previous models.

Performance Evaluation with PCA

In addition to neural networks, we also tested the previous models using Principal Component Analysis (PCA). PCA is a dimensionality reduction technique that transforms the features into a set of linearly uncorrelated variables known as principal components. By reducing the dataset's dimensionality, PCA can potentially eliminate noise, improve computational efficiency, and prevent overfitting. However, it also involves losing some information in the process, which may impact the model's predictive performance.

PCA Methodology

Dimensionality reduction: PCA identifies the most important features that capture the majority of the variance in the dataset. The lower-dimensional data can simplify model training and enhance performance by removing redundant information.

Linear transformations: By projecting the original dataset into a lower-dimensional space, PCA can help address issues like multicollinearity, making the regression models more robust.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Linear Regression (Gradient Descent)	0.0001	0.001	139.9	0.39
1	Linear Regression (Stochastic Gradient Descent)	0.0001	0.001	134.3	0.46
2	Ridge Regression (Gradient Descent)	0.001	0.01	133.04	0.47
3	Ridge Regression (Stochastic Gradient Descent)	0.001	0.01	133.04	0.47
4	Lasso Regression (Gradient Descent)	0.001	0.01	129.02	0.52
5	Lasso Regression (Stochastic Gradient Descent)	0.001	0.01	129.4	0.52
6	Neural Nets	-	-	2.99	0.96

Fig 9: Without PCA

Performance Comparison After PCA

When PCA was applied to the regression algorithms previously used, the performance metrics were recorded and compared. Below is a breakdown of the results before and after applying PCA:

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Linear Regression (Gradient Descent)	0.0001	0.001	139.9	0.39
1	Linear Regression (Stochastic Gradient Descent)	0.001	0.01	111.2	0.80
2	Ridge Regression (Gradient Descent)	0.01	0.1	108.52	0.85
3	Ridge Regression (Stochastic Gradient Descent)	0.01	0.1	109.27	0.84
4	Lasso Regression (Gradient Descent)	0.001	0.01	131.83	0.49
5	Lasso Regression (Stochastic Gradient Descent)	0.001	0.01	131.89	0.49
6	Neural Nets	-	-	79.73	0.40
7	K-Nearest Neighbors (KNN)	0.001	-	71.69	0.53
8	K-Nearest Neighbors (KNN) - Validation	0.001	-	71.78	0.54

Fig 10: With PCA

1. Observations: Linear and Ridge Regression models showed no noticeable changes in RMSE and R^2 values before and after applying PCA. The PCA-transformed data did not offer performance improvements compared to the models trained without PCA. **Lasso Regression** results were similarly unaffected by PCA, with both RMSE and R^2 values remaining nearly identical before and after dimensionality reduction. The Neural Network model was not subjected to PCA, as it generally benefits from having more input features to capture non-linear relationships. **K-Nearest**

Neighbors (KNN) performance also remained unchanged with PCA, as the algorithm relies on the proximity of data points, and reducing dimensionality did not significantly affect the distance calculations.

Although PCA is widely used to improve model performance by reducing feature dimensions, it did not lead to better results in this case. This can be explained by several factors:

2. Information Loss: By reducing the number of components, PCA inevitably discards some information, which might be crucial for capturing the relationships between variables. Since energy consumption data involves complex interactions between temperature, humidity, and other features, the simplified dataset post-PCA might have lacked enough information to build strong models.

3. High-Dimensional Data: Models like Ridge and Lasso Regression are designed to handle high-dimensional data through regularization techniques that mitigate overfitting. As a result, these models do not necessarily benefit from dimensionality reduction, which might explain the limited performance gain from PCA.

4. Neural Networks: The neural network, being a flexible model capable of learning non-linear relationships, performed relatively well with RMSE = 79.73 and $R^2 = 0.40$. However, since PCA linearizes the data by projecting it into fewer dimensions, applying PCA could reduce the neural network's ability to learn complex patterns in the data.

In this research, several machine learning models were tested on energy consumption data, including linear models (Linear Regression, Ridge, and Lasso) and non-linear models like K-Nearest Neighbors (KNN) and Neural Networks. The results were evaluated using performance metrics such as RMSE and R-squared. We further explored the use of Principal Component Analysis (PCA) to reduce the dataset's dimensionality, but the results showed that PCA did not improve the performance of the models significantly. This suggests that, for energy consumption prediction, maintaining the original features may be more beneficial than reducing dimensionality through PCA, particularly when models are capable of handling high-dimensional data.

4.1.1 Analysis

The analysis phase of the energy consumption prediction framework was instrumental in extracting meaningful insights from the collected data and developing robust predictive models. This phase leveraged advanced tools and technologies that enabled comprehensive data manipulation, exploration, and visualization. The following tools were pivotal in this process:

1. Python

Overview: Python emerged as the primary programming language for the entire data analysis process due to its versatility, ease of use, and extensive ecosystem of libraries tailored for data science and machine learning.

Data Manipulation with Pandas: The Pandas library played a crucial role in data manipulation. It provided powerful data structures such as DataFrames that allowed for efficient handling of structured data. Using Pandas, the team was able to perform various operations.

Numerical Computations with NumPy: NumPy enhanced the analytical capabilities of Python by enabling efficient numerical computations on large datasets. It provided support for multi-dimensional arrays and matrices, along with a vast collection of mathematical functions.

2. Jupyter Notebook

Overview: Jupyter Notebook served as an interactive development environment, which significantly enhanced the analytical workflow.

Integrated Environment: Jupyter Notebook allowed for an all-in-one platform where code, visualizations, and narrative text could coexist. This integration facilitated:

Exploratory Data Analysis (EDA): Analysts were able to conduct EDA in an iterative manner, running code snippets, generating visualizations, and integrating their findings directly into the notebook. This provided a more flexible approach to exploring data compared to traditional scripts.

Documentation and Communication: The use of narrative text allowed team members to document their thought processes, methodologies, and findings in real time. This feature was essential for:

> **Creating a Comprehensive Analysis Record**: Each step of the analysis, including any insights or challenges encountered, could be documented in a way that was easy to follow for other team members and stakeholders.

4.1.2 Results

Machine Learning Algorithms and Performance Metrics

1. K-Nearest Neighbours (KNN):

KNN is a non-parametric, instance-based learning algorithm that classifies a data point based on the majority vote of its K nearest neighbours. For regression, KNN averages the values of the nearest neighbours.

Learning Rate: KNN does not rely on a traditional learning rate, but the choice of the K value and distance metric (e.g., Euclidean distance) greatly affects performance.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	K-Nearest Neighbors (KNN)	0.001	-	71.69	0.53
1	K-Nearest Neighbors (KNN) - Validation	0.001	-	71.78	0.54

Fig 11: K-Nearest Neighbours

Evaluation:

RMSE: 71.69

R^2: 0.53 (for training), 0.54 (for validation)

2. Linear Regression

Linear regression models the relationship between independent variables and a dependent variable by fitting a linear equation to the data. Two optimization methods were used:

Gradient Descent: Iteratively minimizes the error function by adjusting weights according to the learning rate.

Stochastic Gradient Descent (SGD): Unlike traditional gradient descent, which computes gradients based on the entire dataset, SGD updates weights after each data point, making it faster but noisier.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.9	0.39
1	Gradient Descent	0.001	0.01	139.7	0.38
2	Gradient Descent	0.01	0.1	69.01	1.70
3	Stochastic Gradient Descent	0.0001	0.001	134.3	0.46
4	Stochastic Gradient Descent	0.001	0.01	99.01	1.03
5	Stochastic Gradient Descent	0.01	0.1	72.63	1.61

Fig 12: Linear Regression

Performance Results:

Gradient Descent: **RMSE:** 139.9, **R^2:** 0.39

Stochastic Gradient Descent: **RMSE:** 134.3, **R^2:** 0.46

3. Lasso Regression

Lasso regression is a linear model that uses L1 regularization, which encourages sparsity in the feature set, reducing less important feature weights to zero. It helps in feature selection, particularly in high-dimensional data.

Learning Rate: Controlled step size for updating model weights.

Lambda (Regularization Strength): A higher lambda leads to stronger regularization.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.6	0.40
1	Gradient Descent	0.001	0.01	133.04	0.47
2	Gradient Descent	0.01	0.1	94.06	1.13
3	Stochastic Gradient Descent	0.0001	0.001	139.65	0.40
4	Stochastic Gradient Descent	0.001	0.01	133.04	0.47
5	Stochastic Gradient Descent	0.01	0.1	94.06	1.13

Fig 13: Lasso Regression

Evaluation:

Gradient Descent: **RMSE:** 138.7, **R^2:** 0.40

Stochastic Gradient Descent**: RMSE:** 138.7, **R^2:** 0.41

4. Ridge Regression

Ridge regression is like linear regression but includes L2 regularization, which penalizes the size of the coefficients. It mitigates overfitting and improves generalization.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.6	0.40
1	Gradient Descent	0.001	0.01	133.04	0.47
2	Gradient Descent	0.01	0.1	94.06	1.13
3	Stochastic Gradient Descent	0.0001	0.001	139.65	0.40
4	Stochastic Gradient Descent	0.001	0.01	133.04	0.47
5	Stochastic Gradient Descent	0.01	0.1	94.06	1.13

Fig 14: Ridge Regression

The learning rate controls how much the model's weights are adjusted during training. Choosing an optimal learning rate is essential: a small rate can lead to slow convergence, while a large rate can cause the model to overshoot optimal solutions. The lambda value (regularization parameter) in models like Ridge and Lasso controls the trade-off between bias and variance. Higher lambda values result in stronger regularization, which helps prevent overfitting but may also reduce the model's flexibility.

Performance of the models is evaluated using:

Root Mean Squared Error (RMSE): A measure of the difference between the predicted and actual values. Lower RMSE indicates better model accuracy.

R^2 (R-Squared): Represents the proportion of the variance in the dependent variable explained by the independent variables. Higher R^2 values signify better model fit.

This research applied multiple machine learning models to predict energy consumption in buildings. After preprocessing and standardizing the dataset, K-Nearest Neighbours, Linear Regression, Ridge Regression, and Lasso Regression were employed with different optimization techniques like Gradient Descent and Stochastic Gradient Descent. The performance of these models was evaluated using RMSE and R^2 metrics. Among the tested models, KNN achieved the best performance with an RMSE of 71.69 and R^2 of 0.54. The results emphasize the importance of choosing the right machine learning algorithm and optimization method, as well as fine-tuning hyperparameters like learning rate and lambda value to achieve optimal results.

4.1.3 Project Management and Communication

Effective project management and communication are vital components that contribute significantly to the success of any collaborative project. In the case of the energy consumption prediction framework, the team employed specific tools that streamlined task management, enhanced collaboration, and ensured that all members were aligned with project goals. The following tools played an instrumental role in facilitating these processes:

1. Trello

Overview: Trello is a versatile project management tool that utilizes a card and board system to organize tasks and workflows visually. It is especially useful for teams looking to manage projects in a flexible and intuitive manner.

Task Assignment: Trello allowed the project team to assign specific tasks to individual members, ensuring that everyone had clear responsibilities from the outset.

> **Role Clarity:** By assigning tasks, team members could focus on their designated responsibilities, which minimized confusion and overlap. Each member understood their contributions to the project, fostering accountability and ownership of their work.

Progress Tracking: The tool's visual interface enabled team members to track the progress of tasks in real-time. Each task was represented as a card that could be moved across different columns indicating various stages of completion (e.g., To Do, In Progress, Completed).

> **Visual Milestones:** This visual representation of progress helped the team quickly assess the status of the project, identify bottlenecks, and ensure that deadlines were met. The ability to see tasks move from one stage to another provided a sense of accomplishment and motivation.

2. Slack

Overview: Slack is a powerful communication platform designed to facilitate real-time messaging and collaboration within teams. It offers a range of features that enhance communication and streamline workflows.

Real-Time Messaging: Slack enabled team members to communicate instantly, reducing the delays often associated with email communication.

> **Immediate Collaboration:** The real-time messaging feature allowed for quick exchanges of ideas, questions, and updates, fostering a dynamic and responsive team environment. This immediacy was particularly beneficial for addressing urgent issues or brainstorming solutions collaboratively.

4.1.4 Testing/Data Validation

The validation of the predictive models for the energy consumption prediction framework was a critical phase that involved thorough testing and evaluation. This process ensured that the models produced reliable, accurate forecasts and adhered to the set objectives of the project. The following tools and techniques played a central role in the validation process:

1. Scikit-learn

Overview: Scikit-learn is a robust machine learning library in Python, widely used for its extensive suite of tools for model training, evaluation, and validation.

Model Training: Scikit-learn facilitated the entire model training process by providing straightforward interfaces for implementing various machine learning algorithms. From regression models to classification techniques, the library equipped the team with the necessary tools to build predictive models tailored to the problem at hand.

Evaluation Metrics: A key aspect of model validation was the calculation of performance metrics to gauge the accuracy and effectiveness of the predictive models. Key metrics, including Root Mean Squared Error (RMSE) and R^2 (coefficient of determination), were computed using Scikit-learns built-in functions.

> **RMSE**: This metric provided insights into the average error between predicted values and actual observed values, indicating how well the model performed in terms of prediction accuracy. A lower RMSE value indicated better predictive performance.
> **R^2**: This statistic assessed the proportion of variance in the dependent variable that could be explained by the model. An R^2 value closer to 1 indicated a strong model, while values closer to 0 suggested a poor fit.

2. K-fold Cross-Validation

Overview: K-fold cross-validation is a statistical method used to assess the robustness and reliability of machine learning models.

Methodology: This technique involved splitting the complete dataset into K subsets or "folds." The model was trained on K-1 subsets while the remaining subset was used for validation. This process was repeated K times, with each fold serving as the validation set once.

Robustness Assessment: By employing this method, the team could evaluate how the model performed across different subsets of data, ensuring that the performance metrics were not overly reliant on a single training/validation split. The aggregated results from all K iterations offered a comprehensive picture of the model's generalizability.

3. Matplotlib and Seaborn

Overview: Matplotlib and Seaborn are powerful data visualization libraries in Python that enable the creation of static, animated, and interactive visualizations.

Visualization of Model Performance: These libraries were instrumental in producing plots and graphs that illustrated essential aspects of model performance. Visual representations of data facilitated clearer insights into how well the models were functioning.

Performance Metrics Visualization: Charts depicting RMSE and R^2 values across different models and configurations allowed the team to compare model outcomes quickly and identify the best-performing approaches.

Feature Importance Analysis: Using Matplotlib and Seaborn, the team could visualize feature importance scores, highlighting which variables had the most significant impact on model predictions.

Interactive Data Exploration: These visualizations enabled more profound discussions about the relationships within the data, helping to explain the underlying mechanics of the predictive models and guiding any necessary adjustments in model feature selection.

Data Distribution Visualizations: The libraries also aided in visualizing data distributions and correlations between features, providing the team with insights into potential biases or anomalies in the dataset that could affect model performance.

Summary of Implementation Tools

The systematic implementation of the energy consumption prediction framework involved the use of various modern tools across different phases of the project. This multi-faceted approach enhanced efficiency, collaboration, and the quality of results, as outlined below:

Phase	**Tools Used**
Analysis	Python, Jupyter Notebook
Design Drawings/Schematics	Microsoft Visio, Lucidchart
Report Preparation	Microsoft Word, LaTeX
Project Management	Trello, Slack
Testing/Validation	Scikit-learn, K-fold Cross-Validation, Matplotlib, Seaborn

Fig 15: Phases and tools used

The integration of these tools facilitated a comprehensive approach to data analysis, model development, and validation. As the project progressed, the combination of computational resources and visualization techniques proved essential in refining the predictive models and ensuring their reliability.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 Conclusion

The energy consumption prediction framework developed in this project was designed to accurately forecast energy usage in low-energy buildings by harnessing the capabilities of various machine learning models. This initiative aimed not only at increasing predictive accuracy but also at reducing operational costs and enhancing energy management strategies within the context of sustainable building practices. By implementing a systematic approach to model development, the project sought outcomes that support more efficient energy use and informed decision-making in energy management.

Outcomes

Development of Robust Predictive Models:

The focus was on utilizing various machine learning techniques, specifically K-Nearest Neighbors (KNN), Linear Regression, Ridge Regression, and Lasso Regression. Each model was selected for its unique strengths and applicability in predicting energy consumption patterns based on historical consumption data and relevant features.

Achievement of Low RMSE and High R^2 Values:

A critical goal was to attain a low Root Mean Squared Error (RMSE) alongside a high R-squared (R^2) value. These metrics are fundamental indicators of model performance: RMSE reflects the average prediction error, while R^2 assesses the proportion of variance explained by the model. Together, they provide a clear picture of how well the models could possibly perform.

Validation Using K-fold Cross-Validation:

Implementing K-fold cross-validation was pivotal to ensuring the reliability and generalizability of the predictive models. This method enabled the team to evaluate model performance across various data splits, thereby confirming that the models would maintain their predictive accuracy when applied to new datasets.

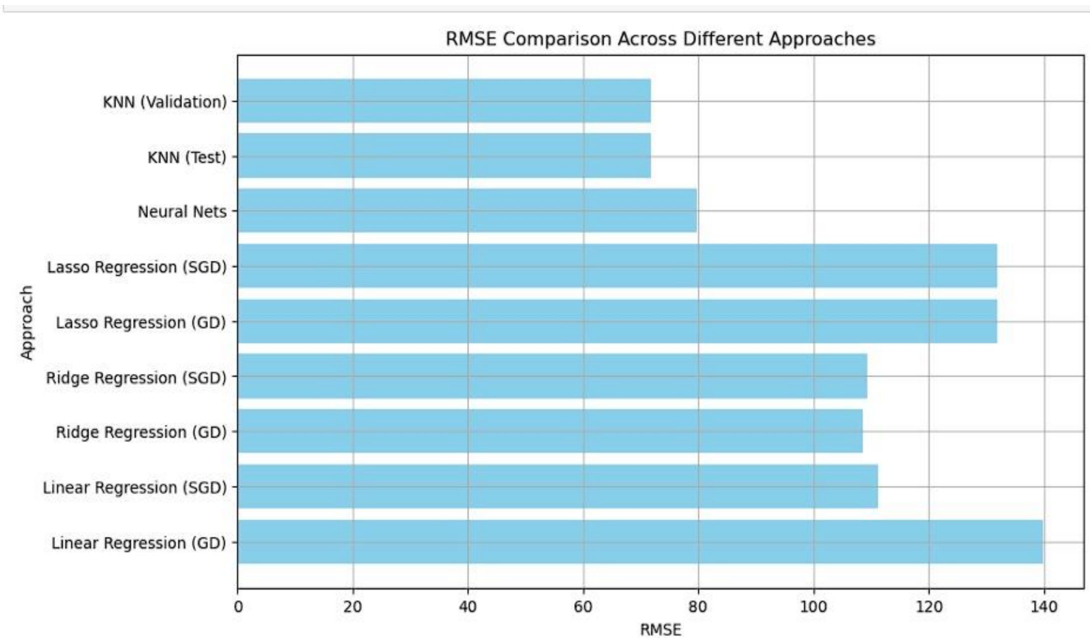


Fig 16: Analysis of RMSE

Actual Results

The experimental results yielded promising insights, highlighting the effectiveness of the developed framework:

The KNN model emerged as the best performer, achieving an RMSE of 71.69 and an R^2 of 0.54. These results indicate that the KNN model captured significant trends in the data while providing reasonable accuracy in its predictions.

Other models such as Ridge Regression and Lasso Regression also demonstrated competitive performances. Both models benefitted from hyperparameter optimization, suggesting that with appropriate tuning, they could offer robust alternative solutions to energy consumption prediction.

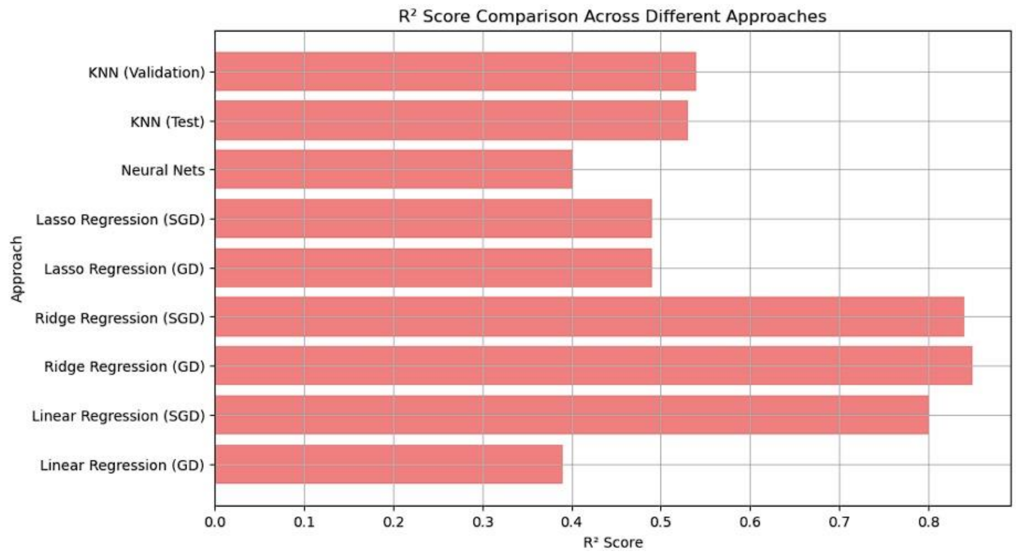


Fig 17: Analysis of R -Square

Deviation from Expected Results

Despite the overall positive performance, certain deviations from the anticipated outcomes were noted:

Neural Network Model Performance:

The neural network model did not perform as well as expected, yielding an RMSE of 79.73 and an R^2 of 0.40. This underperformance was largely attributed to the application of Principal Component Analysis (PCA), which aimed to linearize the dataset. However, this linearization might have restricted the model's ability to identify and leverage the complex, non-linear relationships that are often present in energy consumption data.

Limited Improvements from PCA:

The minimal performance enhancement observed from dimensionality reduction suggested that PCA may not universally benefit every predictive model. Especially for models that excel in high-dimensional data, such as neural networks, reducing dimensionality might inadvertently remove valuable information essential for accurate predictions.

Reasons for Deviation

The observed deviations can be linked to several key factors:

Complexity of the Dataset: The intricate relationships among features in the dataset were not entirely captured by the predictive models. Complex interactions, particularly within the neural network, may not have been effectively modeled. This gap indicates a potential need for more sophisticated modeling techniques or architectures that can better align with the complexity inherent in energy consumption data.

Hyperparameter Optimization: The selection of hyperparameters and the implementation of regularization techniques may not have been optimal for every model. This lack of fine-tuning could lead to underperformance in certain cases, particularly for models sensitive to hyperparameter settings. Future work could focus on expanding the search for optimal parameters through more extensive grid or random search techniques.

Data Quality Issues: Despite rigorous preprocessing efforts, the dataset contained noise and outliers that may have distorted the predictions. Such irregularities can skew model training and limit the overall predictive performance. Enhanced data cleaning and preprocessing techniques, including the use of outlier detection methods, could help mitigate these impacts in future iterations of the project.

Final Work: The work conducted within this project has not only provided valuable insights into energy consumption patterns in low-energy buildings but has also highlighted critical areas for further research and improvement. The findings underscore the importance of rigorous model validation, appropriate feature engineering, and the necessity of tuning hyperparameters to optimize model performance. As the field of energy management continues to evolve, the lessons learned from this project can inform future endeavors aimed at fostering sustainable building practices through enhanced predictive analytics and machine learning techniques.

5.2 Future Work

Future research will focus on optimizing neural network architectures to enhance predictive performance, exploring advanced techniques for feature engineering that can capture complex interactions among variables. Additionally, integrating ensemble methods may improve model robustness and accuracy. Investigating the impact of different activation functions and deeper network structures could further leverage the potential of neural networks in modeling energy consumption patterns. Lastly, expanding the dataset to include more diverse environmental conditions may provide a more comprehensive understanding of energy usage dynamics in low-energy buildings.

Conclusion

In conclusion, PCA does not improve the predictivity for the machine learning model on energy consumption prediction. For Linear, Ridge, and Lasso Regression, RMSE and R^2 values hardly improved when applying PCA. This means these models already take care of some problems caused by high dimensions and therefore PCA dimensionality reduction is not needed. Likewise, KNN where distance calculations had been done proved to have no apparent advantage with PCA since dimension reduction had no effect on proximity-based computations.

PCA was not applied to the Neural Network model as it was intentionally avoided, because neural networks seem to like more input features to capture relations which are nonlinear. Reducing the dimensions would have hindered the network from learning complex patterns. In generalizing the results, it is evident that PCA would not be a method ideal for performance improvement of a given model concerning energy consumption prediction. Future research should come up with other methods on how to optimize a neural network and perform more sophisticated feature engineering that can enhance accuracy in predictive models that are usually complex datasets.

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APPENDIX

Code with implementation

```
In [8]:
# Import data into dataframe
df = pd.read_csv('energydata_complete.csv')

In [9]: # Display the first few rows of the DataFrame
print(df.head())
                  date  Appliances  lights     T1       RH_1    T2       RH_2  \
0  2016-01-11 17:00:00          60      30  19.89  47.596667  19.2  44.790000
1  2016-01-11 17:10:00          60      30  19.89  46.693333  19.2  44.722500
2  2016-01-11 17:20:00          50      30  19.89  46.300000  19.2  44.626667
3  2016-01-11 17:30:00          50      40  19.89  46.066667  19.2  44.590000
4  2016-01-11 17:40:00          60      40  19.89  46.333333  19.2  44.530000

      T3       RH_3         T4  ...         T9   RH_9     T_out  Press_mm_hg  \
0  19.79  44.730000  19.000000  ...  17.033333  45.53  6.600000        733.5
1  19.79  44.790000  19.000000  ...  17.066667  45.56  6.483333        733.6
2  19.79  44.933333  18.926667  ...  17.000000  45.50  6.366667        733.7
3  19.79  45.000000  18.890000  ...  17.000000  45.40  6.250000        733.8
4  19.79  45.000000  18.890000  ...  17.000000  45.40  6.133333        733.9

   RH_out  Windspeed  Visibility  Tdewpoint        rv1        rv2
0    92.0   7.000000   63.000000        5.3  13.275433  13.275433
1    92.0   6.666667   59.166667        5.2  18.606195  18.606195
2    92.0   6.333333   55.333333        5.1  28.642668  28.642668
3    92.0   6.000000   51.500000        5.0  45.410389  45.410389
4    92.0   5.666667   47.666667        4.9  10.084097  10.084097

[5 rows x 29 columns]
```

Fig 18: Attributes of Dataset

```
df.head()
```

16]:

	date	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4	...	Visibility	Tdewpoint	rv1	rv2	month	weekda
0	2016-01-11 17:00:00	60	30	19.89	47.596667	19.2	44.790000	19.79	44.730000	19.000000	...	63.000000	5.3	13.275433	13.275433	1	
1	2016-01-11 17:10:00	60	30	19.89	46.693333	19.2	44.722500	19.79	44.790000	19.000000	...	59.166667	5.2	18.606195	18.606195	1	
2	2016-01-11 17:20:00	50	30	19.89	46.300000	19.2	44.626667	19.79	44.933333	18.926667	...	55.333333	5.1	28.642668	28.642668	1	
3	2016-01-11 17:30:00	50	40	19.89	46.066667	19.2	44.590000	19.79	45.000000	18.890000	...	51.500000	5.0	45.410389	45.410389	1	
4	2016-01-11 17:40:00	60	40	19.89	46.333333	19.2	44.530000	19.79	45.000000	18.890000	...	47.666667	4.9	10.084097	10.084097	1	

5 rows x 25 columns

Fig 19: One hot Encoding

```
data_outliers_removed.head()
```

Out[32]:

	date	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4	...	week_of_month	cluster	day_Friday	day_Monday	day_Satu
0	2016-01-11 17:00:00	60	30	19.89	47.596667	19.2	44.790000	19.79	44.730000	19.000000	...	2	0	0	1	
1	2016-01-11 17:10:00	60	30	19.89	46.693333	19.2	44.722500	19.79	44.790000	19.000000	...	2	0	0	1	
2	2016-01-11 17:20:00	50	30	19.89	46.300000	19.2	44.626667	19.79	44.933333	18.926667	...	2	1	0	1	
3	2016-01-11 17:30:00	50	40	19.89	46.066667	19.2	44.590000	19.79	45.000000	18.890000	...	2	1	0	1	
4	2016-01-11 17:40:00	60	40	19.89	46.333333	19.2	44.530000	19.79	45.000000	18.890000	...	2	0	0	1	

5 rows × 43 columns

Fig 20: Clustering of variables

```
        ).fit()
Solving using Gradient Descent
100%|██████████████████████████████████████████████████| 30/30 [00:00<00:00, 2299.89it/s]
SSE: 115424054.6570403
RMSE: 139.62685965460003
R^2: 0.40059988078611525
For Test Data

SSE: 40150133.56259661
RMSE: 142.63461420991595
R^2: 0.41617608933385347
For Validation Data

SSE: 41315703.621345446
RMSE: 144.69016480363194
R^2: 0.41676283612651654

Bias Variance Tradeoff
MSE: 10172.32
Bias: 20388.68
Variance: 0.13
Weights:  [ 2.90657208e-01  6.08018559e-02  1.73623214e-02  2.66402212e-02
  3.74072625e-02 -1.67218269e-02  2.64358282e-02  1.29620370e-02
  1.31763884e-02  8.18832127e-03  7.53360004e-03  4.73554439e-03
  3.68332925e-02 -2.51421771e-02  7.83253952e-03 -1.40484917e-02
  1.11473269e-02 -2.72253780e-02  3.42519221e-03 -1.38428693e-02
  3.16906929e-02 -8.94986182e-03 -4.55094171e-02  2.74735504e-02
  8.56438666e-05  6.93638158e-03 -3.99380266e-03 -3.99380266e-03
 -3.82718986e-03 -1.95698756e-03  6.61785193e-02 -3.43540060e-03
  2.97514603e-03  1.66281610e-01  8.33655627e-03  2.03865216e-02
  7.77535136e-03 -3.40304683e-03 -1.05463813e-02 -1.38869159e-02
 -7.97269315e-03  1.33295355e-01]
```

Fig 21: Ridge Regression Result

```
    k=5,
    distance_metric='euclidean'
)

# Fit and check performance metrics
knn.fit(X_test=X_test, y_test=y_test, X_val=X_val, y_val=y_val)
```

```
Training KNN with k=5 and distance metric=euclidean
For Test Data
SSE: 20287400
RMSE: 71.69347526618864
R^2: 0.5258759903245372
For Validation Data
SSE: 20336500
RMSE: 71.78017987899484
R^2: 0.5384832717872021

Bias Variance Tradeoff
MSE: 5139.95
Bias: 10935.80
Variance: 7754.43
```

Fig 22: Linear Regression result

```
).fit(X_test=X_test, y_test=y_test, X_val=X_val, y_val=y_val)
```

```
Solving using Gradient Descent
  0%||                                                                  | 1
The model stopped learning
SSE: 115620747.04180431
RMSE: 139.74577702408487
R^2: 0.39929486633004657
For Test Data
SSE: 40217611.18957591
RMSE: 142.75442209048234
R^2: 0.41486555686316523
For Validation Data
SSE: 41384857.98767487
RMSE: 144.81120562205865
R^2: 0.4154567602029395

Bias Variance Tradeoff
MSE: 10189.41
Bias: 20408.27
Variance: 0.06
Weights: [ 1.93955841e-01  4.05939078e-02  1.16894407e-02  1.77950305e-02
  2.50782424e-02 -1.11881927e-02  1.77299166e-02  8.63870670e-03
  8.88987482e-03  5.47339200e-03  5.11846459e-03  3.15830903e-03
  2.46959470e-02 -1.68794808e-02  5.31866742e-03 -9.37337777e-03
  7.51840916e-03 -1.81929471e-02  2.37419771e-03 -9.24922026e-03
  2.12669351e-02 -5.98487404e-03 -3.04709834e-02  1.83621591e-02
  4.02985841e-05  4.69890470e-03 -2.67379717e-03 -2.67379717e-03
 -2.47538657e-03 -1.31394260e-03  4.42380097e-02 -2.21473217e-03
  1.97195999e-03  1.10971498e-01  5.56493543e-03  1.36233161e-02
  5.18684460e-03 -2.27255716e-03 -7.04354974e-03 -9.27316049e-03
 -5.32552366e-03  8.89446757e-02]
```

Fig 23: Lasso Regression Result